



UNITED STATES MILITARY ACADEMY
WEST POINT

Chemical Engineering

Course and Program AAR (AY26-2)

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21 May 2026



Agenda

- ☐ **CH367**
- ☐ **CH362**
- ☐ **CH364**
- ☐ **CH400**
- ☐ **CH402**
- ☐ **CH300**
- ☐ **CH450**
- ☐ **Chemical Engineering Program AAR
Comments**



Sustain:

- ✓ **Capstone with CH364.**
- ✓ **Emphasis on fundamentals, like modeling, correlating processes to block diagrams, and P&IDs.**
- ✓ **Daily quizzes. Consider doing these at the beginning of class, no resources allowed. With the integration of AI into all that we do, the Cadets are using it to answer questions on the quizzes. We could better test their comprehension of the reading by doing it in class and it could still be done on Canvas.**
- ✓ **Continue use of SSI for describing/demonstrating dynamic behavior, stability, controller design, and tuning. The Cadets get a lot out of SSI and it allows you to start relatively simple and scale up.**

Improve:

- ✓ **One problem should be added to the TEE to increase breadth slightly; recommend a tuning/stability problem.**
- ✓ **Introduce python and ChemCAD to model controllers and control loops. This was accomplished, but still needs work. Start the process of upgrading the textbook to support this.**
- ✓ **Improve the controls portion of the Capstone. Improve format/quality of block diagrams and P&IDs, and in general provide more guidance.**
- ✓ **Consider integrating SSI into a few homework problems; guided simulation and analysis.**



Course average: **83.8%** (AY25-2: 83%); TEE average **75%** (AY25-2: 70%)

Sustains

- ❑ Maintain majority of points from in class assignments (74% this year)
 - ❑ Largely maintain overall structure – but make time for cadets to brief the PS solutions and/or add short in-class concept quizzes on Canvas
- ❑ Maintain hands-on lab and optimization labs
- ❑ Performance on **TEE 75% increased (71% in AY25; 1's & 0's)** - bonus point oral briefs of PS 8 and WPR 3 corrections, and a TEE review lesson all assisted cadets in tying concepts together
- ❑ Provide old graded events for cadets to use to practice; cadets studied much more using these old graded events and it levelled the playing field

Improves

- ❑ Split Capstone into individual writing assignment and group modelling project
 - ❑ Individual writing assignment: cadets dive deeply into a topic that interests them
 - ❑ Group modelling project: through a video tour or trip section, whole class learns about a ChemE process – they model it in groups and then experiment in ChemCAD to optimize the system – this would introduce them to design and optimization
- ❑ Add more oral briefs for points – some cadets could do the math, but didn't understand the concepts, and vice-versa; 3 trials of oral briefs this year all yielded substantial development
- ❑ Return WPRs to 55-minute lessons – 120-minute WPRs did not lead to better results or understanding
 - ❑ Use additional lab time to experiment in ChemCAD, as cadets never grasped the power of the software or how a ChemE would use it
- ❑ Poor Performance on **WPR3 67% (88% in AY25; 1's & 0's)** - cadets struggled with tying M&E balances together in a short time period
- ❑ Emphasize self-study using learncheme.com
- ❑ Consider developing a new TEE – current TEE emphasizes PV work 4 times



Section III AY26-2 Course Assessment

- ❑ Course average: **88.4%** (AY25-2: 86.7%); TEE average **82.5%** (AY25-2: 81.9%)

Sustains

- ❑ Combined capstone project for both CH364/CH367.
- ❑ Dedicated Lessons and Lab periods as capstone working sessions.
 - ❑ Capstone **IPR performance improved to 88% (1's & 0's)**
- ❑ Performance on **WPR2 (88%) and WPR3 (84.5%) increased (1's & 0's)**.
- ❑ Improved performance on Selectivity with multiple rxns **Problem C on TEE 93% (83% in AY25; 1's & 0's)**

Improves

- ❑ Average on Problem A FEE Style Questions on **TEE declined 69% (AY25-2: 72.2%; from 1's & 0's)**.
 - ❑ Add Multiple choice FEE questions to WPRs
- ❑ Tie in BH sub-basement Bath and CSTR reactor experiments to labs 3 & 4.
- ❑ Poor performance on P&IDs on capstone.
 - ❑ Use Capstone Working Periods to model P&ID using LucidChart on Canvas
- ❑ Introduce cost estimation in CHEMCAD for reactor sizing with each lab.
- ❑ Develop additional PS problems, with increased reliance on numerical solvers and multiple concepts.



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Sustain:

- ❑ 19/25 Cadets Passed the FEE Overall. **76.0% Pass Rate**
- ❑ **No 2nd Attempts**
 - ❑ **6 Failures Total: Faherty, Leja, Ebiuwhe, Macune, Stokes, Tringali**
- ❑ 2 lessons per topic: CRE, Thermo, Controls, Transport, Chem/Bio, Seps
 - ❑ 1st lesson is practice problems & FE Ref Manual, 2nd lesson is Quiz
- ❑ Used Canvas Quizzes as timed quizzes (25 min) with FEE references
 - ❑ Cadets came to class and took a “topic FEE” with immediate results
 - ❑ Instructor and cadets received immediate feedback on performance
 - ❑ Instructor worked specific problems with remainder of class time
- ❑ Continue SSI modules: DIST & ACU
- ❑ 2x practice WPRs and full WPR; good recap of all topics in FEE format.

Improve:

- ❑ Trim question banks in PPI modules for more focused problem sets
- ❑ Re-write the Ethics quiz; questions nuanced / not directly from FEE manual
- ❑ Add additional SSI-Simulator Exercises
 - ❑ ACU Exercise with scenarios and/or CTOW with Level Controller



Sustain:

Capstone project - Use AIChE design problem if possible; recent problems are good fit with curriculum, but not this year. If possible, use multiple problems.

Time survey quizzes, design groups based on grades, and field trip were good ideas.

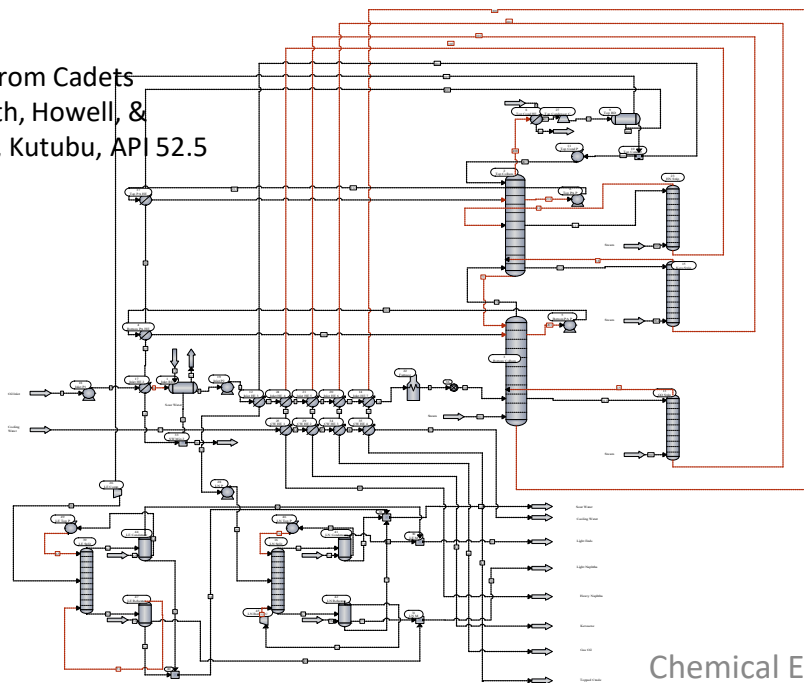
Improve:

Control – Knowledge of control schemes and ability to draw P&IDs could be better. Some groups started early but some did not make sense (49% safety and 59% controls compared to 73% and 70% last AY, from 1/0's)

Reports – Cadets need a report template. LyX taught in CH365.

Research – Special topics research was poor 57% vs 95% last AY.

CC PFD from Cadets
Mantooth, Howell, &
Ferrying, Kutubu, API 52.5



Research:

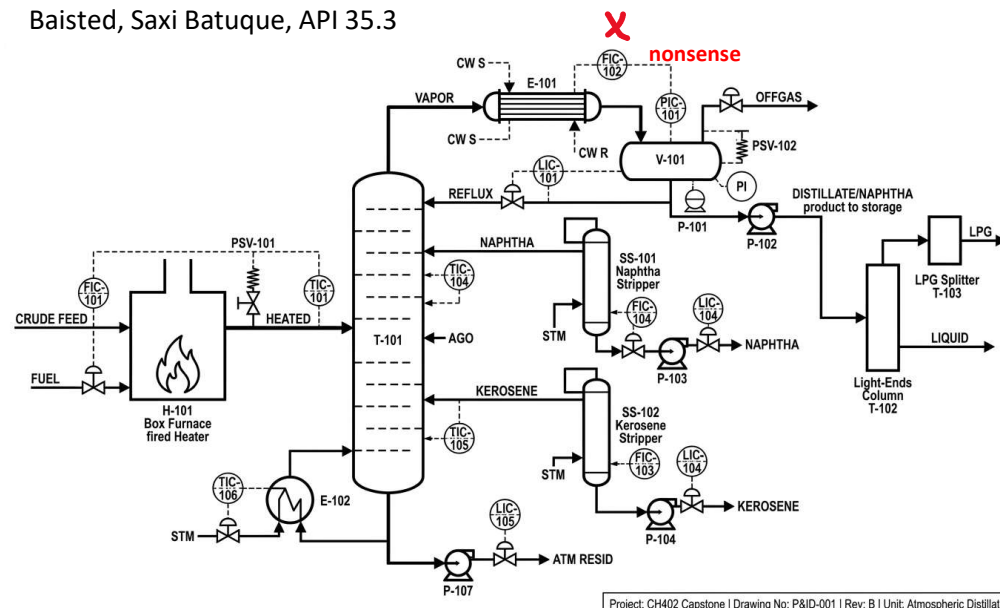
Discussion of boiling point analysis for all groups was not strong. This is a big improvement from last year.

Research on sulfur and water content to define design specs was good.

Research on context and social issues was good.

Research on special topics was generally poor. Suggest changing the topic by giving them a pic list of crude oils and asking them to design a process PLUS a feed cocktail that enables them to meet BP specs for JP8 instead of kerosene.

P&ID from Cadets Thomas, Faherty, &
Baisted, Saxi Batuke, API 35.3



Project: CH402 Capstone | Drawing No: P&ID-001 | Rev: B | Unit: Atmospheric Distillation.



- ❑ **Course average: 91.9%; TEE average 88.8%**

Sustain:

- ✓ Continue using in-class activities to guide students through formulating governing equations and deriving quantitative results.
- ✓ Capstone project on the bio scaffold has been shown as the most favorite graded component. Cadets displayed good engagement, particularly when exploring practical biomedical applications (**AY25-2: 94.7%; AY26-2: 94.6%**).

Improve:

- ✓ Align in-class coding exercises with contemporary subject matter to keep the material relevant (i.e., mRNA vaccine distribution/automated insulin pumps can be related to ODEs).
- ✓ Broaden the problem-solving exercises to emphasize numerical solvers and cross-concept integration.
- ✓ Improve the application problems in the Term-End Exam (TEE) to better assess cadets' ability to apply bioengineering fundamentals in practical contexts (**AY25-2: 87.2%; AY26-2: 88.8%**).



❑ **Course average: 92.1%**

Sustain:

- ✓ **Continue incorporating in-class activities to illustrate the formulation of governing equations and the process of obtaining quantitative results since the class population is diverse (AY26-2: 10 CEN, 7 BIO, 1 EVE).**
- ✓ **Maintain the use of end-of-block quizzes to truly assess cadets' comprehension of key learning objectives:**
 - AY25-2: Quiz 1: 100.0%, Quiz 2: 95.2%, Quiz 3: 97.6%, Quiz 4/5: 100.0%**
 - AY26-2: Quiz 1: 92.6%, Quiz 2: 89.4%, Quiz 3: 89.0%, Quiz 4: 91.1%, Quiz 5: 92.2%**

Improve:

- ✓ **Expand the in-class coding exercises with the contemporary subject matters (i.e., in-depth focus on pharmacokinetic models or biosignaling).**
- ✓ **With the advent of generative A.I., having the oral examination as part of either end-of-semester Captone Project or TEE would be beneficial to assess the cadets' fundamental understanding of bioengineering principles.**



1. **ABET e-Coursebooks/Documents will be priority in Summer**
 - ❑ **On-Site Visit on 30AUG-02SEP26**
 - ❑ **All Course Assessments Scrub (Dr. Nagelli & Dr. Yuk)**
 - **CH363 & CH383 from AY26-1**
 - ❑ **AY26-2 CD's work on e-Coursebooks**
 - ❑ **Faculty Meetings with Program Evaluator (31AUG)**
2. **ABET Lab Safety (POC: Dr. Yuk with with LTC Jerke, Safety OIC)**
 - ❑ **SB51/53 Lab (Clean up, SDS, Fire extinguisher, Fume hood)**
 - ❑ **CH101, CH383, CH275 Labs**
3. **Course Directors Handbook for all administrative guidance and SOP**
 - ❑ **EXSUM on Dept Sharepoint to include Course AAR Slides**
 - ❑ **Course Folder on Sharepoint (Rosters – alpha & OML, Course Assessment, all Course Materials (SIS, lesson schedule/objectives, grades etc.)**
 - ❑ **Graded Events Folder on Sharepoint (all graded events, solutions, and cut scales (labs, quizzes, WPRs, TEE, capstone, etc.)**
 - ❑ **Course Assessment Due 30 days after semester completion**



4. CH459 Summer Training (Dr. Yuk + **Junior Faculty Members?**)

- ☐ Steam generator for EVAP
- ☐ Running all unit operations
- ☐ Lab safety and cleaning for ABET Walkthrough

5. CH485 Summer FDW (LTC Plante + Dr. Nagelli)

- ☐ LTC Plante is CD for CH485 in AY27-1
- ☐ Run all labs for CH485 and practice lessons with senior faculty in attendance (Prof. Biaglow, Dr Nagelli, and Dr. Yuk)

6. Conferences

- ☐ **AICHE Fall 2026** in Minneapolis, MN (NOV26) – Undergrad poster competition for cadets to compete (1 poster per cadet; S: SEP 2026) (POC: CPT Glinski as OIC)
 - ☐ Faculty & Grad Oral Talk Abstracts Submitted: 1LT Eddie Chen, Dr. Yuk, & Dr. Johnson

7. Program Led “events”

- ☐ Chili Cookoff (OCT/NOV26, OIC: MAJ Frey, AOICs: MAJ Rogers, CPTs Stewart & Glinski)

8. Calendar Events

- ☐ 29MAY: Dept On-Site (Conf Rm; APs, Assoc Prof and higher)
Topics:
 - ☐ *Current Major Requirements*
 - ☐ *Minimum Major Requirements*
 - ☐ *ABET Assessment Requirements*
 - ☐ *Lab Consolidation Plan/Tax?*
 - ☐ *Department Events: AAR/Reassess*
- ☐ 21JUN: Incoming Faculty Report Date
- ☐ 22-26JUN: New Faculty In-processing
- ☐ 29JUN: Class of 2030 R-Day
- ☐ 29JUN-24JUL: FDW
- ☐ 6-27JUL: CH101 STAP3



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ABET Self-Study & On-Site Visit Timeline

Vist Dates: 30AUG-02SEP26



Engineering
Accreditation
Commission

Accredited to September 30, 2027



Computing
Accreditation
Commission

Accredited to September 30, 2027



Applied and
Natural Science
Accreditation
Commission

Accredited to September 30, 2032

Next General Review

Accredit to September 30, 2027. A request to ABET by January 31, 2026 will be required to initiate a reaccreditation evaluation visit. In preparation for the visit, a Self-Study Report must be submitted to ABET by July 1, 2026. The reaccreditation evaluation will be a comprehensive general review.



Record Year Timeline

2026-2027 Criteria available

ABET Board Meeting – approval of
2026-2027 criteria

Submit Request for Evaluation (RFE)
to ABET – 31 January 2026

ABET Commission Meeting:
ABET's Institutional Representatives
Day

Self-Studies Submitted to
ABET – **1 July 2026**

2025

2026

August

September

October

November

December

January

February

March

April

May

June

July

15 January

Self-Study Preparation

Round-Robin Review of
Self-Studies

Final Draft of Self-Studies Complete –
15 May 2026

Documentation Collection

ABET Record Year



- Approve TC (USMA Level)
- Approve PEVs when assigned (USMA Level)
- Finalize Self-Studies – check for completeness and prepare for submission
- Submit Self-Studies
- Finalize all ET credit review memos with AARs and ensure updated Redbook Entries
- Finalize and complete external access site for display materials
- Complete adjustments to web pages
- Brief new senior leadership on ABET and ensure visit is scheduled on calendars
- Verify fire extinguishers and any other lab safety is updated
- Receive common e-Coursebooks for display from core courses
- Complete Logistics
 - Schedules for Team, TC, PEVs
 - Ground Transportation on-post
 - Tour of West Point (Jenn Voigtschild)
 - Legal Review for Tues lunch
 - Planning for Lunch at Club Monday
 - Complete purchase process to pay for the visit
- Positive communication with team prior to visit



- Complete Self-Studies
- Complete record-year assessments and data collection
- Organize all policy memos etc.
- Organize your display materials and be prepared to put onto site for outside access
- Positive communication with PEVs (primary and alternates throughout the summer)
- USMA will provide transcripts on request
- Prepare your Faculty (all faculty and leadership)
- Prepare your lab staff
- Prepare your students
- Reach out to Advisory Board Members who may be able to come on Monday Lunch